

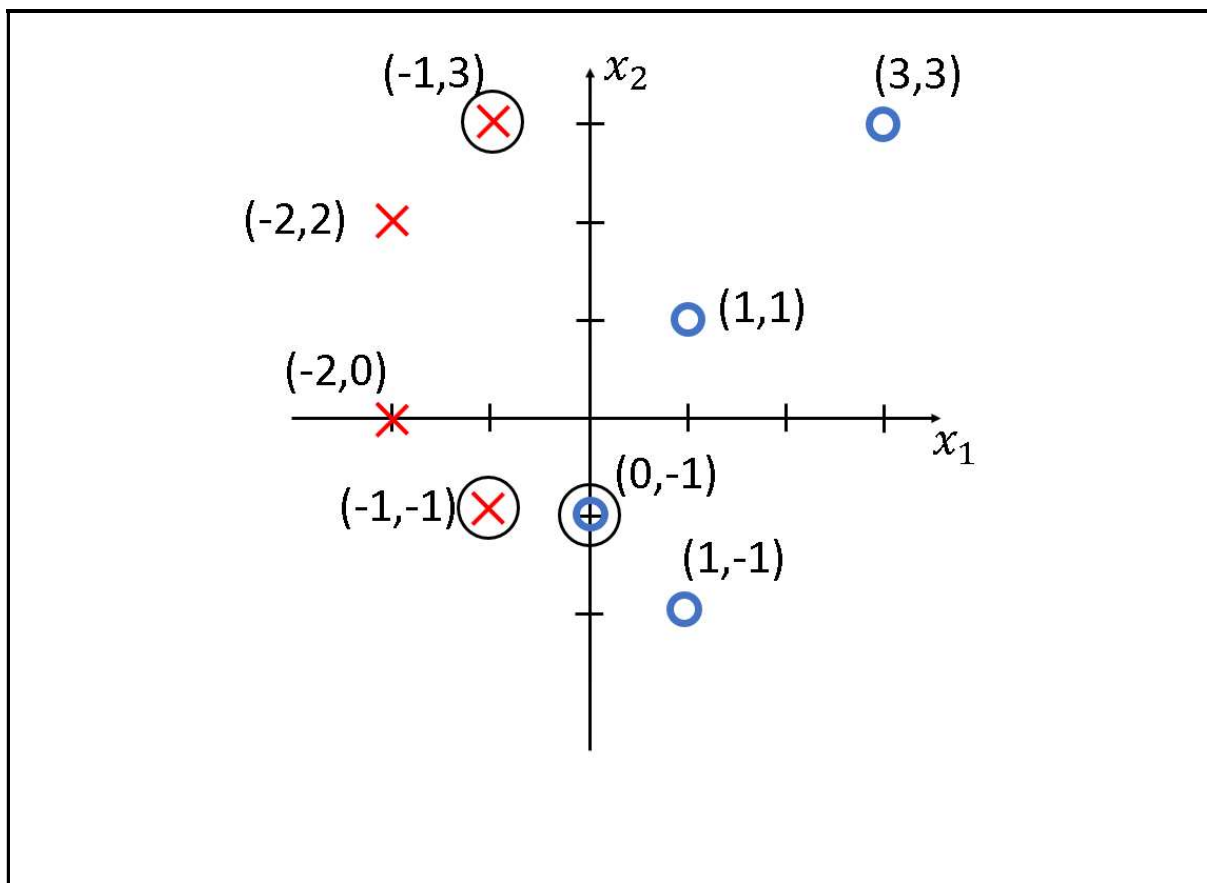
Question 1A What is the “kernel trick”?

[3 marks]

A kernel is a function that maps the training set into a different space where the data points might then become linearly separable. In lectures, we mentioned the use of this technique in both polynomial regression and for SVM.

Question 1B Circle the support vector(s).

[3 marks]



Question 1C Alternative PCA Formulation.

[4 marks]

The covariance matrix $\mathbf{X}^T \mathbf{X}$ is of size $n \times n$, while the data matrix $\mathbf{X}$ is of size $m \times n$. Which formulation is preferred depends on the relative sizes of m and n because it impacts the time taken for the SVD computation. Note that there's a cost to computing $\mathbf{X}^T \mathbf{X}$ as well.

Question 1D Speeding up Gradient Descent.

[4 marks]

Many possibilities including:

- Perform feature scaling and mean normalization on the training data.
- Apply PCA to reduce the dimensionality of the training data, but taking care to keep enough variance, i.e. at least 99%.
- Use a larger learning rate or an adaptive learning rate to speed up convergence.
- If the dataset is too large, possibly run stochastic gradient descent or mini-batch gradient descent instead for faster iterations.

Question 2A Entropy of the COVID outcomes

[3 marks]

The outcomes counts for Positive, Negative and Other are 5, 5, and 2.

$$\begin{aligned}\text{Entropy} &= -\frac{5}{12} \log\left(\frac{5}{12}\right) - \frac{5}{12} \log\left(\frac{5}{12}\right) - \frac{2}{12} \log\left(\frac{2}{12}\right) \\ &= 0.526 + 0.526 + 0.431 \\ &= 1.483\end{aligned}$$

Question 2B Decision tree to decide if patient has COVID-Rho.

[15 marks]

We just need to consider the remaining entropy.

$$\begin{aligned}
 \text{Remainder} &= \sum_{i=1}^v \frac{p_i + n_i + o_i}{p + n + o} I\left(\frac{p_i}{p_i + n_i + o_i}, \frac{n_i}{p_i + n_i + o_i}, \frac{o_i}{p_i + n_i + o_i}\right) \\
 \text{Remainder(Fever)} &= \frac{6}{12} I\left(\frac{2}{6}, \frac{2}{6}, \frac{2}{6}\right) + \frac{6}{12} I\left(\frac{3}{6}, \frac{3}{6}, 0\right) \\
 &= -\frac{1}{2} \left(\frac{1}{3} \log_2\left(\frac{1}{3}\right) \times 3 \right) - \frac{1}{2} \left(\frac{1}{2} \log_2\left(\frac{1}{2}\right) \times 2 \right) \\
 &= 1.292 \\
 \text{Remainder(Cough)} &= \frac{6}{12} I\left(\frac{2}{6}, \frac{3}{6}, \frac{1}{6}\right) + \frac{6}{12} I\left(\frac{3}{6}, \frac{2}{6}, \frac{1}{6}\right) \\
 &= -\frac{1}{2} \left(\frac{1}{3} \log_2\left(\frac{1}{3}\right) + \frac{1}{6} \log_2\left(\frac{1}{6}\right) + \frac{1}{2} \log_2\left(\frac{1}{2}\right) \right) \times 2 \\
 &= 1.459 \\
 \text{Remainder(Tiredness)} &= \frac{7}{12} I\left(\frac{5}{7}, \frac{2}{7}, 0\right) + \frac{5}{12} I\left(\frac{3}{5}, \frac{2}{5}, 0\right) \\
 &= -\frac{7}{12} \left(\frac{5}{7} \log_2\left(\frac{5}{7}\right) + \frac{2}{7} \log_2\left(\frac{2}{7}\right) \right) - \frac{5}{12} \left(\frac{3}{5} \log_2\left(\frac{3}{5}\right) + \frac{2}{5} \log_2\left(\frac{2}{5}\right) \right) \\
 &= 0.908 \\
 \text{Remainder(No Taste)} &= \frac{7}{12} I\left(\frac{4}{7}, \frac{1}{7}, \frac{2}{7}\right) + \frac{5}{12} I\left(\frac{1}{5}, \frac{4}{5}, 0\right) \\
 &= -\frac{7}{12} \left(\frac{4}{7} \log_2\left(\frac{5}{7}\right) + \frac{1}{7} \log_2\left(\frac{1}{7}\right) + \frac{2}{7} \log_2\left(\frac{2}{7}\right) \right) \\
 &\quad - \frac{5}{12} \left(\frac{1}{5} \log_2\left(\frac{1}{5}\right) + \frac{4}{5} \log_2\left(\frac{4}{5}\right) \right) \\
 &= 1.105
 \end{aligned}$$

Hence, the root node should be “Tiredness?”.

Question 2B Decision tree to decide if patient has COVID-Rho (continued). [15 marks]

Thereafter, we have the following sample points remaining for “Tiredness? Yes”:

Patient	Fever?	Cough?	No Taste?	Rho?
A	Yes	No	No	Negative
B	No	Yes	Yes	Positive
E	Yes	No	Yes	Positive
G	No	Yes	No	Positive
H	No	Yes	No	Negative
I	Yes	No	Yes	Positive
M	No	No	Yes	Positive

Again, we just need to consider the remaining entropy.

$$\begin{aligned}
 \text{Remainder(Fever)} &= \frac{3}{7}I\left(\frac{1}{3}, \frac{2}{3}\right) + \frac{4}{7}I\left(\frac{1}{4}, \frac{3}{4}\right) \\
 &= -\frac{3}{7}\left(\frac{1}{3}\log_2\left(\frac{1}{3}\right) + \frac{2}{3}\log_2\left(\frac{2}{3}\right)\right) - \frac{4}{7}\left(\frac{1}{4}\log_2\left(\frac{1}{4}\right) + \frac{3}{4}\log_2\left(\frac{3}{4}\right)\right) \\
 &= 0.857 \\
 \text{Remainder(Cough)} &= \frac{3}{7}I\left(\frac{1}{3}, \frac{2}{3}\right) + \frac{4}{7}I\left(\frac{1}{4}, \frac{3}{4}\right) \\
 &= -\frac{3}{7}\left(\frac{1}{3}\log_2\left(\frac{1}{3}\right) + \frac{2}{3}\log_2\left(\frac{2}{3}\right)\right) - \frac{4}{7}\left(\frac{1}{4}\log_2\left(\frac{1}{4}\right) + \frac{3}{4}\log_2\left(\frac{3}{4}\right)\right) \\
 &= 0.857 \\
 \text{Remainder(No Taste)} &= \frac{4}{7}I\left(\frac{4}{4}, 0\right) + \frac{3}{7}I\left(\frac{1}{3}, \frac{2}{3}\right) \\
 &= -\frac{4}{7}\left(\frac{4}{4}\log_2\left(\frac{4}{4}\right)\right) - \frac{3}{7}\left(\frac{1}{3}\log_2\left(\frac{1}{3}\right) + \frac{2}{3}\log_2\left(\frac{2}{3}\right)\right) \\
 &= 0.394
 \end{aligned}$$

Hence, the next attribute should be “No Taste?”.

Next, we have the following sample points remaining for “Tiredness? No”:

Patient	Fever?	Cough?	No Taste?	Rho?
C	Yes	Yes	Yes	Other
D	Yes	No	Yes	Other
F	Yes	Yes	No	Negative
K	No	Yes	Yes	Negative
L	No	No	No	Negative

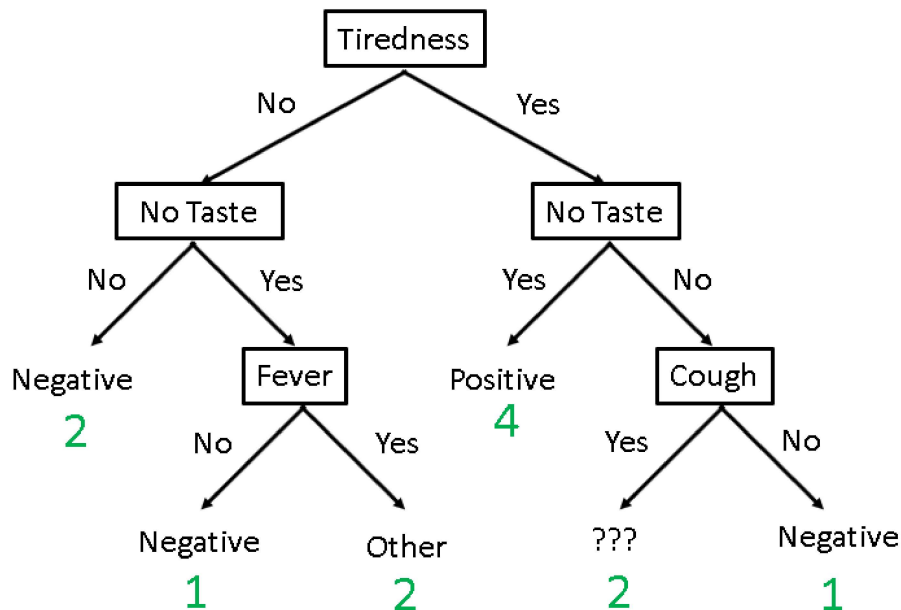
Question 2B Decision tree to decide if patient has COVID-Rho (continued). [15 marks]

Again, we just need to consider the remaining entropy.

$$\begin{aligned}
 \text{Remainder(Fever)} &= \frac{3}{5}I\left(\frac{1}{3}, \frac{2}{3}\right) + \frac{2}{5}I(0, 1) \\
 &= -\frac{3}{5}\left(\frac{1}{3}\log_2\left(\frac{1}{3}\right) + \frac{2}{3}\log_2\left(\frac{2}{3}\right)\right) \\
 &= 0.551 \\
 \text{Remainder(Cough)} &= \frac{3}{5}I\left(\frac{2}{3}, \frac{1}{3}\right) + \frac{2}{5}I\left(\frac{1}{2}, \frac{1}{2}\right) \\
 &= -\frac{3}{5}\left(\frac{1}{3}\log_2\left(\frac{1}{3}\right) + \frac{2}{3}\log_2\left(\frac{2}{3}\right)\right) - \frac{2}{5}\left(\frac{1}{2}\log_2\left(\frac{1}{2}\right) \times 2\right) \\
 &= 0.951 \\
 \text{Remainder(No Taste)} &= \frac{3}{5}I\left(\frac{1}{3}, \frac{2}{3}\right) + \frac{2}{5}I(1, 0) \\
 &= -\frac{3}{5}\left(\frac{1}{3}\log_2\left(\frac{1}{3}\right) + \frac{2}{3}\log_2\left(\frac{2}{3}\right)\right) \\
 &= 0.551
 \end{aligned}$$

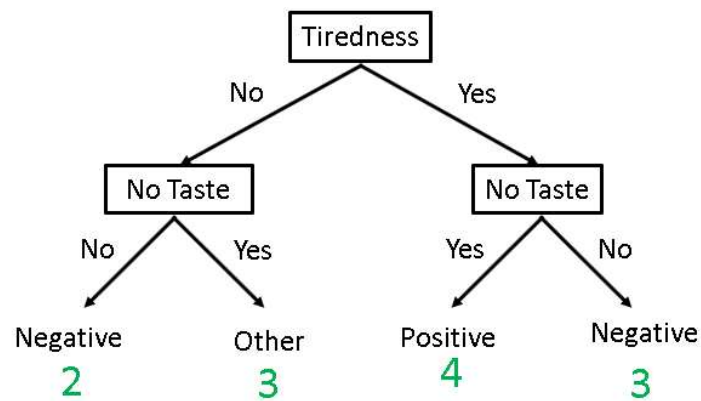
Hence, the next attribute can either be “No Taste?” or “Fever,” but you are expected to break ties in favour of “No Taste?”.

We can deduce the last attribute by inspection and derive the following decision tree:



Question 2C Pruned decision tree (min-leaf ≥ 2).

[4 marks]

**Question 2D** Precision, recall and F1 value.

[6 marks]

The prediction for patient 4 is wrong. It is a false negative.

	Pred -ve	Pred +ve
Actual -ve	$TN = 7$	$FP = 0$
Actual +ve	$FN = 1$	$TP = 2$

$$\begin{aligned}
 \text{Precision} &= \frac{TP}{TP + FP} \\
 &= \frac{2}{2 + 0} \\
 &= 1
 \end{aligned}$$

$$\begin{aligned}
 \text{Recall} &= \frac{TP}{TP + FN} \\
 &= \frac{2}{2 + 1} \\
 &= \frac{2}{3}
 \end{aligned}$$

$$\begin{aligned}
 F1 &= \frac{2}{\frac{1}{P} + \frac{1}{R}} \\
 &= \frac{4}{5}
 \end{aligned}$$

Question 2E Choice between precision and recall.

[3 marks]

If COVID-Rho is very dangerous, it can be potentially fatal if we false to detect it accurate. We want to avoid false negatives, so we want to maximize *recall*.

Question 3A Warm Up: Forward Propagation

[2 marks]

$$\begin{aligned}\hat{y} &= g^{[2]}(\mathbf{W}^{[2]T}(g^{[1]}(\mathbf{W}^{[1]T}\mathbf{x}))) \\ &= \sigma(\mathbf{W}^{[2]T}(\sigma(\mathbf{W}^{[1]T}\mathbf{x})))\end{aligned}$$

Question 3B Find $\delta^{[2]}$.

[2 marks]

$$\begin{aligned}\delta^{[2]} &= \frac{\partial \hat{y}}{\partial \mathbf{f}^{[2]}} \\ &= \left[\sigma(\mathbf{f}^{[2]})(1 - \sigma(\mathbf{f}^{[2]})) \right], \text{ since } \hat{y} = \sigma(\mathbf{f}^{[2]})\end{aligned}$$

Question 3C Find $\frac{\partial \hat{y}}{\partial \mathbf{W}^{[2]}}$.

[2 marks]

$$\begin{aligned}\frac{\partial \hat{y}}{\partial \mathbf{W}^{[2]}} &= \frac{\partial \mathbf{f}^{[2]}}{\partial \mathbf{W}^{[2]}} \frac{\partial \hat{y}}{\partial \mathbf{f}^{[2]}} \\ &= \mathbf{a}^{[1]} \sigma(\mathbf{f}^{[2]})(1 - \sigma(\mathbf{f}^{[2]})), \text{ since } \mathbf{f}^{[2]} = (\mathbf{W}^{[2]})^T \mathbf{a}^{[1]}\end{aligned}$$

Question 3D Find $\frac{\partial \hat{y}}{\partial \mathbf{w}^{[2]}}$ (again).

[2 marks]

$$\frac{\partial \hat{y}}{\partial \mathbf{w}^{[2]}} = \mathbf{a}^{[1]} \boldsymbol{\delta}^{[2]}$$

Question 3E Find $\boldsymbol{\delta}^{[1]}$.

[3 marks]

$$\begin{aligned} \boldsymbol{\delta}^{[1]} &= \frac{\partial \hat{y}}{\partial \mathbf{f}^{[1]}} \\ &= \left(\frac{\partial \mathbf{a}^{[1]}}{\partial \mathbf{f}^{[1]}} \right)^T \frac{\partial \mathbf{f}^{[2]}}{\partial \mathbf{a}^{[1]}} \frac{\partial \hat{y}}{\partial \mathbf{f}^{[2]}} \\ &= \left(\frac{\partial \mathbf{a}^{[1]}}{\partial \mathbf{f}^{[1]}} \right)^T \mathbf{w}^{[2]} \boldsymbol{\delta}^{[2]}, \text{ since } \mathbf{f}^{[2]} = (\mathbf{w}^{[2]})^T \mathbf{a}^{[1]} \\ &= \begin{bmatrix} \frac{\partial a_1^{[1]}}{\partial f_1^{[1]}} & \frac{\partial a_1^{[1]}}{\partial f_2^{[1]}} \\ \frac{\partial a_2^{[1]}}{\partial f_1^{[1]}} & \frac{\partial a_2^{[1]}}{\partial f_2^{[1]}} \end{bmatrix}^T \mathbf{w}^{[2]} \boldsymbol{\delta}^{[2]}, \text{ since } \mathbf{a}^{[1]} = \sigma(\mathbf{f}^{[1]}) \\ &= \begin{bmatrix} \sigma(f_1^{[1]})(1 - \sigma(f_1^{[1]})) & 0 \\ 0 & \sigma(f_2^{[1]})(1 - \sigma(f_2^{[1]})) \end{bmatrix} \mathbf{w}^{[2]} \boldsymbol{\delta}^{[2]}, \text{ since } \mathbf{a}^{[1]} = \sigma(\mathbf{f}^{[1]}) \end{aligned}$$

Since the sigmoid function is element-wise, $\sigma(f^{[1]})(1 - \sigma(f^{[1]})) \mathbf{w}^{[2]} \boldsymbol{\delta}^{[2]}$ is also good.

Question 3F Find $\frac{\partial \hat{y}}{\partial \mathbf{w}^{[1]}}$.

[3 marks]

$$\begin{aligned} \frac{\partial \hat{y}}{\partial \mathbf{w}^{[1]}} &= \frac{\partial \mathbf{f}^{[1]}}{\partial \mathbf{w}^{[1]}} \left(\left(\frac{\partial \mathbf{a}^{[1]}}{\partial \mathbf{f}^{[1]}} \right)^T \frac{\partial \mathbf{f}^{[2]}}{\partial \mathbf{a}^{[1]}} \frac{\partial \hat{y}}{\partial \mathbf{f}^{[2]}} \right)^T \\ &= \frac{\partial \mathbf{f}^{[1]}}{\partial \mathbf{w}^{[1]}} (\boldsymbol{\delta}^{[1]})^T \\ &= \mathbf{x}^{[0]} (\boldsymbol{\delta}^{[1]})^T, \text{ since } \mathbf{f}^{[1]} = (\mathbf{w}^{[1]})^T \mathbf{x}^{[0]} \end{aligned}$$

Question 3G Compute $\frac{\partial \hat{y}}{\partial \mathbf{w}^{[1]}}$ and $\frac{\partial \hat{y}}{\partial \mathbf{w}^{[2]}}$ (to 3 significant digits).

[6 marks]

$$\begin{aligned}
 \mathbf{f}^{[1]} &= (\mathbf{W}^{[1]})^T \mathbf{x}^{[0]} \\
 &= \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\
 \mathbf{a}^{[1]} &= \sigma(\mathbf{f}^{[1]}) \\
 &= \begin{bmatrix} \sigma(3) \\ \sigma(1) \end{bmatrix} = \begin{bmatrix} 0.953 \\ 0.731 \end{bmatrix} \\
 \mathbf{f}^{[2]} &= (\mathbf{W}^{[2]})^T \mathbf{a}^{[1]} \\
 &= \begin{bmatrix} 3 & 4 \end{bmatrix} \begin{bmatrix} \sigma(3) \\ \sigma(1) \end{bmatrix} \\
 &= \begin{bmatrix} 3\sigma(3) + 4\sigma(1) \end{bmatrix} = \begin{bmatrix} 5.781 \end{bmatrix} \\
 \mathbf{a}^{[2]} &= \sigma(\mathbf{f}^{[2]}) \\
 &= \begin{bmatrix} \sigma(5.781) \end{bmatrix} \\
 \delta^{[2]} &= \begin{bmatrix} \sigma(5.781)(1 - \sigma(5.781)) \end{bmatrix} = \begin{bmatrix} 0.00306 \end{bmatrix} \\
 \frac{\partial \hat{y}}{\partial \mathbf{W}^{[2]}} &= \sigma(\mathbf{f}^{[2]})(1 - \sigma(\mathbf{f}^{[2]}))\mathbf{a}^{[1]} \\
 &= 0.00306 \begin{bmatrix} \sigma(3) \\ \sigma(1) \end{bmatrix} = \begin{bmatrix} 0.00291 \\ 0.00224 \end{bmatrix} \\
 \delta^{[1]} &= 0.00306 \begin{bmatrix} \sigma(3)(1 - \sigma(3)) & 0 \\ 0 & \sigma(1)(1 - \sigma(1)) \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} \\
 &= \begin{bmatrix} 0.000415 \\ 0.00241 \end{bmatrix} \\
 \frac{\partial \hat{y}}{\partial \mathbf{W}^{[1]}} &= \mathbf{x}^{[0]}(\delta^{[1]})^T \\
 &= \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 0.000415 & 0.00241 \end{bmatrix} \\
 &= \begin{bmatrix} 0.000415 & 0.00241 \\ 0.000415 & 0.00241 \end{bmatrix}
 \end{aligned}$$

Notice the vanishing gradients!

Question 4A Suggest a method to select a good value of k .

[4 marks]

- +1 Run k-means on multiple k , and compute the loss for each configuration.
- +3 Plot the loss against the k -values, and apply the elbow method to select a good k .

Question 4B Explain why this random initialization method might not be good. [4 marks]

- +3 By assigning samples to random clusters, each cluster's centroid would be found somewhere close to the mean of the dataset.
 - +1 Good random initializations of centroids usually leverage on the centroids being far away from each other/exploiting locality.
 - +1 Poor initialization of the clusters will increase likelihood of converging to suboptimal local minimas.
- Max 4m, no marks awarded for stating that number of iterations will increase, or that random assignment of points to clusters will result in random final centroids.

Question 4C What is causing poor k-means performance and how to resolve? [4 marks]

- If student mentions any of the following reasons, max 2m altogether:
- 2m Poor initialization of the clusters, run more iterations with different initial centroids.
 - 2m Bad choice of k , use elbow method to select a good k .
 - 2m Noise in the dataset, re-take the data or manual cleaning of the data.

The root cause of poor performance in a large dataset with many features is the Curse of Dimensionality, stating any the following reasons, max 4m altogether:

- 4m Features are different scales (i.e. some values are boolean ranging from 0-1, while others are in the 0-10 range) and will affect the clustering, normalize the data to resolve.
- 4m Large number of features, which means that the model suffers from the curse of dimensionality, run PCA to reduce the number of dimensions.

Marks will be deducted for not stating the problem without a reasonable resolution.

Question 5A Does a full 3x6 kernel behave as a fully connected layer? Explain. [2 marks]

Yes it is similar to a fully connected layer, but without bias terms. Since a fully connected layer has equation $W^T x + b$, then the n -by- n kernel multiplying by the image would behave as $W^T x$, without the bias term, or with $b = 0$.

+1 Explain that each kernel will produce a single output from the entire image, k kernels will produce a k outputs similar to k output neurons.

+1 Note that the bias term is 0 in this CNN case.

Question 5B Value of the bottom-right output pixel? Padding 0, stride 2. [2 marks]

C+I+O-E-K-Q

Note that since we have no padding and a stride of 2, the output is only a 1×2 array, and the column with F, L, R is not reached.

Question 5C Value of the bottom-right output pixel? Padding 1, stride 1. [2 marks]

K+Q

First, pad the original image with a layer of 0 values, then slide the kernel to the bottom-right.

Question 5D Should we use CNN or not? Explain. [3 marks]

Both networks would be able to solve the problem, but CNNs are better suited. CNN kernels **leverage on spatial context** (by only assigning weights to nearby pixels) to extract local features (as opposed capturing global features in a fully connected layer), enjoy **translation invariance**, so features appearing in one part of the image can also be recognized elsewhere, and **have fewer parameters**, as the kernels share weights, which means training is faster.

Mentioning any of the three points is sufficient to get 3m, but justification provided should be stated clearly, otherwise partial marks will be awarded instead.

Question 5E CNN classifier dimensions and number of parameters.

[6 marks]

Layer	Layer dimension	Number of trainable parameters
Input	$28 \times 28 \times 3$	0
Conv ₁	$24 \times 24 \times 4$	$(5 \times 5 \times 3 + 1) \times 4 = 304$
MaxPool ₁	$12 \times 12 \times 4$	0
Conv ₂	$8 \times 8 \times 10$	$(5 \times 5 \times 4 + 1) \times 10 = 1010$
MaxPool ₂	$4 \times 4 \times 10$	0
Linear	160×1	$160 + 1 = 161$

Question 5F Describe your CNN experiment setup.

[5 marks]

1. Split dataset into train and test sets in a 9:1 ratio.
2. Perform image augmentation separately on since 1000 images is too little.
3. Upsample the minority class.
4. Train the classifier on the training set.
5. Score the classifier using F1-score.

Other marks may be awarded or deducted based on reasoning (or lack thereof).

Question 6 Did you see the animal?

[3 marks]

